Phys 224 Problem Set 4

Due Monday Nov. 2 at 11:59 PM. You may talk/consult with whom or whatever, but you must write your own solutions for submission.

1) In the last quiz, we had a freezer where the speed of sound was $100\pi = 314$ metres per second. What temperature was the freezer?

2) In class, someone asked how good the approximation that the *difference* in path lengths is $d\sin(\theta)$ In class, I set the angle θ to be the angle from one speaker to our receiver, but we're better off if we make θ the angle to the midpoint of the two speaker. Consider a concert hall where we have two speakers 5 metres apart, so at (x, y) positions $(-2.5, 0)$ and $(2.5, 0)$. We have a listener 20 m away at a 45 degree ($= \pi/4$ radians) angle, so the position is $(10\sqrt{2}, 10\sqrt{2})$.
Part a) What is the actual difference in distance between the speakers and the listener? Recall that the exact distance between $(x1, y1)$ and $(x2, y2)$ is $\sqrt{(x1-x2)^2 + (y1-y2)^2}$.
Part b) What would we have predicted the difference to be using $d\sin(\theta)$?
Part c) What is the difference in distances in parts a and b? This is the error in our $d\sin(\theta)$ approximation. If all has gone well, it should be of order 1cm.
Part d) At what frequency would this error become important? I will define "important" to be 1 radian of phase. Below this frequency, $d\sin(\theta)$ is a very good estimate, but you might want to be careful at higher frequencies.

3) It turns out you are putting on the concert in the previous problem, and you discover a studio exec is attending and is sitting in the $(10\sqrt{2}, 10\sqrt{2})$ seat. You're hoping they give you a record deal, so you want to give them the best possible sound. You want the sound from the speakers to be in-phase at their seat, which you can do by putting a delay into one of the speakers. How long should that delay be?

4) You go to a concert, but being a poor student, you get a seat right behind a pillar. If the pillar is 1 metre wide, what frequencies will you hear well/what ones will the pillar block?

5) Find a bottle in your house and predict its Helmoltz resonant frequency. Please include your estimates of the volume of the resonator, and the length/area/volume of the neck. Include how you estimated these value. Then go play it and report the actual pitch. How close are you? Use whatever tool you want to measure the pitch - phyphox or other apps on your phone, or compare to an instrument. If you like, you can use audacity on a computer, or if you're feeling extra adventurous, the python script I wrote (plot_spectrum_jls.py in the python directory on github).